Vector-Valued Functions

Section 1: Position, Velocity, and Acceleration Vectors

1 A particle has position vector $\mathbf{r}(t) = \langle t^3 - 2t, \sin(2t) \rangle$. Find the acceleration vector $\mathbf{a}(t)$ at $t = 0$.

a. $\langle 0, 0 \rangle$

b. $\langle -2, 2 \rangle$

c. $\langle 0, -4 \rangle$

d. $\langle 6, 0 \rangle$

2 If the velocity vector of a particle is $\mathbf{v}(t) = \langle 3t^2, 4t \rangle$ and $\mathbf{r}(0) = \langle 1, 5 \rangle$, find $\mathbf{r}(1)$.

a. $\langle 2, 7 \rangle$

b. $\langle 1, 2 \rangle$

c. $\langle 4, 9 \rangle$

d. $\langle 3, 4 \rangle$

3 What is the magnitude of the acceleration vector $\mathbf{a}(t) = \langle 3, -4 \rangle$?

a. 5

b. 7

c. $\sqrt{7}$

d. 1

Section 2: Finding Position from Velocity & Acceleration

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4 Given $\mathbf{a}(t) = \langle 0, -32 \rangle$ with initial velocity $\mathbf{v}(0) = \langle 20, 64 \rangle$, what is $\mathbf{v}(2)$?

a. $\langle 20, 0 \rangle$

b. $\langle 20, -32 \rangle$

c. $\langle 0, 0 \rangle$

d. $\langle 20, 32 \rangle$

5 In the trajectory above, what is the maximum height achieved if initial position is $\mathbf{r}(0) = \langle 0, 0 \rangle$?

a. 64 feet

b. 32 feet

c. 128 feet

d. 16 feet

6 If a particle has $\mathbf{v}(t) = \langle \frac{1}{t+1}, 2e^{2t} \rangle$ and $\mathbf{r}(0) = \langle 3, 4 \rangle$, find the x-component $x(t)$.

a. $x(t) = \ln(t+1) + 3$

b. $x(t) = \ln(t+1)$

c. $x(t) = -\frac{1}{(t+1)^2} + 3$

d. $x(t) = e^{t+1} + 2$

Section 3: Speed, Direction, and Displacement

7 What is the direction angle θ of motion when $\mathbf{v}(t) = \langle -2, 2\sqrt{3} \rangle$?

a. $\frac{2\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{4\pi}{3}$

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8 Which vector represents the net displacement of a particle from $t = 1$ to $t = 4$?

a. $\int_1^4 \mathbf{v}(t)dt$

b. $\int_1^4 \|\mathbf{v}(t)\|dt$

c. $\mathbf{v}(4) - \mathbf{v}(1)$

d. $\mathbf{a}(4) - \mathbf{a}(1)$

9 If speed is constant, what must be true about the acceleration vector and the velocity vector?

a. $\mathbf{a}(t) \cdot \mathbf{v}(t) = 0$ (orthogonal)

b. $\mathbf{a}(t)$ is zero

c. $\mathbf{a}(t)$ is parallel to $\mathbf{v}(t)$

d. $\mathbf{a}(t) = -\mathbf{v}(t)$

Section 4: Distance Traveled with Vector-Valued Functions

10 Write an integral for the total distance traveled by a particle from $t = 0$ to $t = 5$ with $\mathbf{v}(t) = \langle \cos(t^2), e^{\sqrt{t}} \rangle$.

a. $\int_0^5 \sqrt{\cos^2(t^2) + e^{2\sqrt{t}}}\, dt$

b. $\int_0^5 (\cos(t^2) + e^{\sqrt{t}})dt$

c. $\sqrt{\int_0^5 \cos^2(t^2)dt + \int_0^5 e^{2\sqrt{t}}dt}$

d. $\int_0^5 \|\mathbf{a}(t)\|dt$

11 A particle moves with speed $\sqrt{9 + 16t^2}$. What is the total distance traveled from $t = 0$ to $t = 1$?

a. $\int_0^1 \sqrt{9 + 16t^2}\, dt$

b. $\sqrt{25} = 5$

c. $\frac{25}{2}$

d. 4

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12 If a particle starts at $(2, 3)$ and has displacement $\langle -4, 7 \rangle$, what is its final position?

a. $(-2, 10)$

b. $(6, -4)$

c. $(2, 10)$

d. $(-6, 4)$

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